

4

Simple Statistical Techniques for Univariate and Bivariate Analyses

CHAPTER OBJECTIVES
■ Understand the Pearson Product-Moment correlation coefficient and evaluate it for statistical significance. ■ Understand why the Pearson correlation captures linear relationships only, not nonlinear ones, and hence a correlation of zero does not necessarily imply the absence of a relationship. ■ Understand and appreciate what you can vs. cannot conclude from a correlation between variables. ■ How to generate a scatterplot in Python, including a scatterplot matrix revealing univariate as well as bivariate distributions. ■ How to calculate a Spearman correlation as a nonparametric alternative to Pearson. ■ How to compare means using <i>t</i>-tests for independent and paired-samples. ■ How to conduct a binomial test as well as a chi-squared goodness-of-fit test on contingency tables when working with categorical data.

In this chapter, we feature a variety of relatively simple tests that can be used immediately without too much background theory to address many of the research questions you may come across over the course of your career. Though in the most complete and formal sense most of the tests in this chapter have an underlying statistical model, they are nonetheless much easier and straightforward than many of the statistical modeling techniques surveyed in the remainder of the book. For instance, techniques such as the **analysis of variance** (ANOVA) or **regression** involve hypothesizing quite intricate models at times, and require a fair bit of expertise when interpreting such models. In this chapter, however, we do not focus on elaborate statistical models, but instead choose to provide some quick and efficient solutions to address basic research questions that often arise. This does not mean that interpreting such tests in the context of research is always easy. Statistics and its application to empirical data can be very **slippery** in that even what might appear to be a very simple test can be very easily misunderstood and misapplied when interpreting data. Never assume that because a test looks easy, its interpretation in the context of science is equally as simple. Simple tests are often the most misused or misunderstood. There is nothing “simple” about a

t -test if you are unaware of what makes the size of t large or small and conclude something from the test that is incorrect.

Tests in this chapter cover much of the kind that would be covered in introductory statistics at the undergraduate level, and include such things as **correlation**, **non-parametric correlation**, **z-tests for means**, **t-tests**, as well as inferences featuring the use of **categorical** or **count data**, including **binomial tests** as well as **chi-squared goodness-of-fit tests**. The components of these tests form the bedrock of more advanced statistics as well and more complex models.

If you are a true newcomer to statistics, you will have much to learn from these tests. If you are an experienced student or researcher already, all or most of these tests will likely be familiar, yet you are still encouraged to consider each test carefully by embracing the discussion accompanying each one, as there may still be something there for you to learn that may have been missed in previous learning. For instance, tests of correlation are quite simple to conduct. However, what a correlation can actually “tell you” from your data on a scientific level is usually very limited, and a discussion of what correlation does vs. does not provide is essential. Interpretation-wise, **correlation is perhaps one of the most abused statistics computed on data and the most misinterpreted**. Now, that does not mean we have to delve into a deep discussion of the philosophy behind the concept of correlation (that topic could easily fill a chapter in its own right), but it does mean that if one is not at least **familiar** with this discussion, one stands the chance of making quite egregious interpretations of empirical data based on simply computing correlation coefficients! When we conclude something from a statistical test that is methodologically wrong, we are probably better off not computing the statistic in the first place. A false conclusion is often more problematic than no conclusion at all, and such a happening rears its head not only in the social sciences, but also in the natural sciences.

For example, if we found that temperature was correlated with the presence of the COVID-19 virus, and somehow concluded that heat or lack thereof “caused” COVID based only on this correlation, that would be a severe misuse and misunderstanding of what correlation provides or measures, and we would probably do better not computing one at all (if we are to be abusive in the interpretation). In such a case, **our (faulty) interpretation of the statistic is actually misinforming us**. Examples such as this are replete in the research literature where a statistical test is used as backing for a research claim that is simply not supported, and the authors would have done much better not performing or interpreting the test at all. As mentioned earlier in the book, **mediation models** are a prime example of this and of how inaccurate scientific conclusions are drawn based on evidence for statistical (but not necessarily substantive) mediation.

4.1 Pearson Product-Moment Correlation

The **Pearson Product-Moment correlation** is a measure of the linear relationship between variables, and is given by

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1) S_x \cdot S_y} = \frac{\text{cov}_{xy}}{S_x \cdot S_y}$$

The correlation coefficient here is simply the **standardized covariance**. But what does this mean? To understand any statistical function, you need to look at its components. That is, you need to look at the mathematics of it to see what the formula actually “does.” Let us look at the numerator of the quotient:

$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

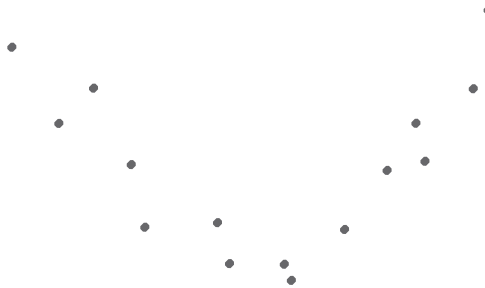
What we have defined immediately above is called the **covariance**. The covariance is the **average cross-product** between variables x_i and y_i . We say “average” cross-product because we are dividing the sum by $n-1$. Recall that an arithmetic average is typically computed by dividing by n , not $n-1$. We are dividing by $n-1$ because we are losing a **degree of freedom**, but nonetheless we notice the quotient is still an “average-like” statistic. The fact that we are losing a degree of freedom does not suddenly make the function “non-average-like.” It is still a kind of average. Covariances are used extensively in statistics and statistical modeling, but they come with a potential problem. The problem is that they are **scale-dependent**, which means they will be influenced by the **raw scale** on which x_i and y_i are computed. If x_i and y_i contain a lot of variability, then all else equal, the sum of products may be quite large, since we are multiplying one deviation by another from respective means. Dividing by $n-1$ does not solve this problem either, since all this is doing is distributing the sum across the $n-1$ pieces of information that went into the sum. Hence, **it is entirely possible to get a large covariance even if the relationship between x_i and y_i is not terribly strong**. This is the essential point.

The solution, as given in Pearson’s r , is to divide by $s_x \cdot s_y$. But why do this? When we divide by $s_x \cdot s_y$, we are **standardizing the covariance**, and hence placing **bounds** on the resulting number between -1.0 and $+1.0$. Why does dividing by $s_x \cdot s_y$ have this effect? While this can be proved mathematically, for applied purposes the proof is not terribly enlightening. Much more useful is to simply look at what the formula is doing. You might think of $s_x \cdot s_y$ as the total product of deviation **possible** in the data. That is, it is the total “cross-product” that is possible for the two measured variables under consideration. In other words, it gives you the “**play space**” on which to evaluate the observed relationship. It is the baseline covariability available to you for the given data. The numerator of the correlation coefficient tells you how much of this “place space” is due to the actual relationship between x_i and y_i and, hence, the **numerator is a subset of this larger play space**. Its value cannot be greater (in absolute value) than the entire play space since the denominator represents the total cross-variation possible. Hence, the correlation coefficient in this regard is accounting for the total cross-variation observed relative to the total cross-variation possible in the given set of data. In this way, the **Pearson correlation coefficient is, in effect, a ratio**. And because we are generating this ratio, the scale of the variables becomes a non-issue, in that the resulting number reflects a **dimensionless relationship** between x_i and y_i . Since the numerator cannot be greater than the denominator, the bounds on the correlation coefficient are necessarily between -1.0 and $+1.0$. The sign of the coefficient will be determined by the sign of the covariance, but the magnitude of the (dimensionless) relationship that exists will be determined by Pearson’s r .

4.2 A Pearson Correlation Does Not (Necessarily) Imply Zero Relationship

Very important to understand is that a **Pearson correlation of zero in the sample does not equate to there being a zero relationship in the data**. This is a mistake often made by students and even experienced researchers, almost equivalent in nature to that of assuming normal distributions by default when discussing any phenomena. For instance, if I told you the mean IQ of the population is 100, you might automatically mentally assume that half of the population has an IQ less than 100 and half has an IQ greater than 100. However, this is incorrect, unless we knew the distribution of IQs to be normally distributed and/or symmetrical (e.g. for a bimodal distribution this statement would be correct also). When we think of “half below” and “half above,” the temptation is usually to automatically incorporate normality. If the distribution is skewed, the statement is false. Of course, for the median the statement is true, but not necessarily for the mean. Another example of how perception is reality, is that if you tell someone shopping for a house that the average home price in the area they are considering is 200k, they might automatically assume that half of the houses are below this price and half above, but this would be an incorrect assumption, given that we do not know the exact form of the distribution. **Always ask about the distribution on which any statistic is computed so you can put that statistic in context of the elements on which it was computed.**

The analogous temptation exists with the correlation coefficient. **A correlation coefficient should never be interpreted without simultaneously visualizing a plot of the two variables.** Now, that is a statement you **should** memorize! Two variables may be related, sometimes strongly so, but not necessarily **linearly**. Pearson’s correlation will only reflect the degree (or non-degree) that variables are linearly related. Its magnitude will not account for non-linear trends. If the relationship is curvilinear, for instance, Pearson’s correlation, though computable on such data, will not reflect the degree of this curvilinearity. Hence, you could have discovered a very strong relationship, yet it would be missed by a simple computation of Pearson’s r . As an example, consider the following plot:



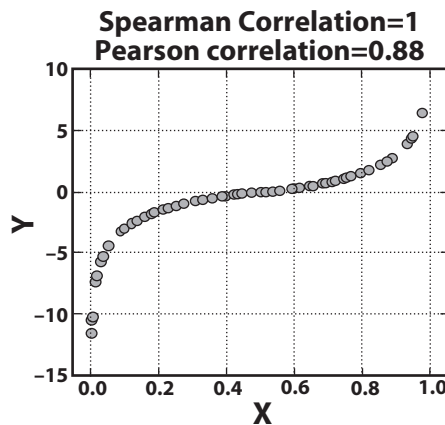
Is there a relationship between the two variables? Absolutely, and quite a strong one at that. However, the relationship is not linear, and for these data at least, the Pearson correlation will actually equal near 0! If one simply computed the correlation via software output without plotting the variables, this potentially important relationship could go unnoticed. **When software reports $r = 0$, it does not necessarily mean variables are unrelated. Again, always plot your data! Never interpret any correlation coefficient (linear or otherwise) without a plot!**



A Pearson Product-Moment correlation assesses the linear relationship among two variables and consists of dividing an average cross-product in the numerator (called the covariance) by the product of standard deviations in the denominator. It is very important to understand that a Pearson r of 0 does not necessarily imply the absence of a relationship between two variables. What it does suggest is the absence of a linear relationship. Always plot your data to potentially detect relationships that are non-linear in form. A coefficient of correlation of any type should never be interpreted without an accompanying plot.

4.3 Spearman's Rho

If the relationship between variables is **non-linear**, one may choose to compute a coefficient that will account for the non-linearity. One coefficient that does this is **Spearman's rho**, named after Charles Spearman who derived the coefficient in 1904. Spearman's rho reflects the relationship between two variables, but is not restricted to that relationship necessarily being linear in form. It may be linear, but it need not be for Spearman's rho to still be quite large. Like Pearson, Spearman's rho ranges from -1.0 to $+1.0$, but it is distinct from that of Pearson in that it measures **monotonically increasing** or **decreasing** relationships, not simply linear ones. What does this mean, exactly? A plot will help illustrate the concept. Consider the following plot:



In this plot note that as x_i increases, y_i also increases, even though the relationship is far from linear. The idea of “monotonically increasing” in this case simply means that each increase in x_i pairs itself with an increase in y_i . As we can see from the plot, the relationship is strong, yet Pearson's correlation would not capture this strength. However, according to Spearman, the relationship is perfect and equal to 1.0 because of the monotonicity. It would be a shame to compute a Pearson correlation on such data without a plot and conclude that you have less of a relationship than actually exists. Again, I repeat (I think this is the third time!) – **ALWAYS PLOT YOUR DATA.**

To compare and contrast a Spearman correlation with a Pearson correlation, consider the following data on two generic variables (we are making up the numbers):

```
x = [0, 2, 6, 7, 15]
y = [0, 1, 8, 13, 20]
df = pd.DataFrame(x, y)
df
```

```
Out[26]:
```

```
0    0
1    2
8    6
13   7
20  15
```

We first compute Pearson r to see what we get. For this, we will use **scipy**, specifically `scipy.stats()`:

```
import scipy.stats
scipy.stats.pearsonr(x,y)
```

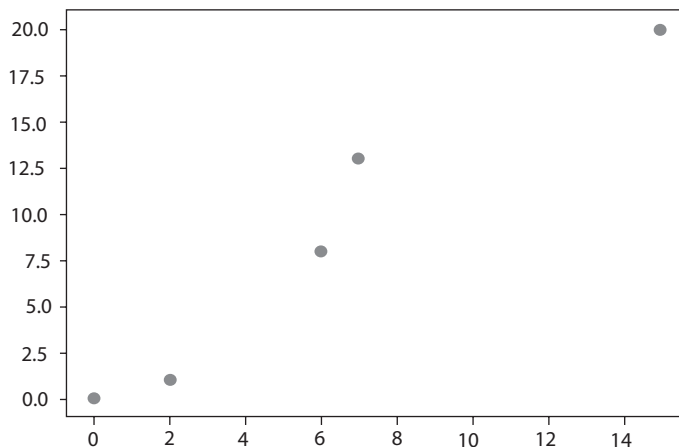
```
Out[51]: (0.973647485990732, 0.0051149659365227395)
```

We can see that the correlation between x and y is equal to 0.97 and is statistically significant with a p -value of 0.005. That it is statistically significant is not the issue for now. The size of it is, however, and we see that it is quite large and impressive. Now, let us try a Spearman rho on the same data:

```
scipy.stats.spearmanr(x,y)
Out[52]: SpearmanrResult(correlation=0.9999999999999999,
pvalue=1.4042654220543672e-24)
```

We can see that Spearman's rho is equal to 1.0 (rounded up), also statistically significant. But why is Spearman a perfect correlation and Pearson not? Look at the data once more. Notice that for every increase in x , there is a corresponding increase in y . This is exactly what Spearman's measures, **the monotonic increase in one variable as a function of another**, regardless of whether or not the relationship is linear. With Pearson, it assesses linearity, and hence the correlation will be less. A plot of the data will help shed light on why Spearman is higher than Pearson here:

```
plt.scatter(x, y)
Out[53]: <matplotlib.collections.PathCollection at 0x1c184978>
```



Notice from this plot that the relationship is not perfectly linear. However, an increase in one variable is consistently accompanied by an increase in the other. This is precisely why Spearman and Pearson do not agree. Pearson tries to essentially fit a line through the scatter, which of course does not fit very well since the scatter is a bit messy. Spearman, on the other hand, attempts to assess whether an increase in x goes with an increase in y , which for these data it does so perfectly. Hence, this is why Spearman is larger than Pearson. Which correlation reflects the “true” relationship, Pearson or Spearman? The question makes no sense, since each coefficient is defined differently, and so it assesses different things. It is not a matter of one being a population coefficient and the other not. Both are sample statistics and both can be used to estimate their respective population counterparts. That is not what distinguishes the correlation coefficients here. What does is what we have emphasized: one assesses linearity, the other a monotonically changing relationship. Hence, this highlights an important universal principle when interpreting statistics in general, which is, **what is revealed depends on what is being measured**. Is there a relationship between x and y ? The only correct immediate answer to this question is, “Depends on what you mean by ‘relationship’, in the Spearman sense, yes, quite strong, in the Pearson sense, slightly less so.” Interpretations of data are only as meaningful as the functions used to compute them. This is why understanding the formulas and equations is paramount, long before you apply them to data.

As we will see when we study regression later in the book, the same question can be raised as it concerns prediction. “Does x predict y ?” It depends on how you define, mathematically, the “idea” of prediction. As we will see, we will usually define it in the sense of least-squares. The “it depends” part is important here. A relationship (or any statistical entity) is only as meaningful as the computation for it. Is there a relationship between COVID-19 and age? It depends on what we are computing to attempt to capture and assess that relationship. The concept of “relationship” is quite hollow without simultaneously including details of the mathematical function used in measuring such a relationship. Formulas and mathematical functions in general tell us first and foremost what we can vs. not conclude about data. Empirical relationships exist only because we have a concept (a mathematical formula) to define them. Otherwise, we cannot measure them. Before correlation was invented by the likes of Francis Galton and company in the 1880s, the idea of a relationship between variables was quite vague. Correlation and regression gave humanity a way of measuring and assessing this “concept.” Again, concepts give rise to statistics and mathematics, and these latter are simply precise and very rigorous ways of getting a handle on the underlying concept. Look at the formula for correlation once more and you will see a symbolic representation of a more elusive concept, that of “relationship.” However, why stop with simply one way of assessing “relationship?” This is exactly why Spearman’s rho was invented, among many other correlation coefficients. **There are different ways of tapping into similar ideas, and in statistics it is no different. Concepts in statistics are philosophical “fuzzy” entities that have been germinating throughout history, until they were eventually defined “rigorously” via symbols (i.e. mathematics).**

4.4 More General Comments on Correlation: Don't Let a Correlation Impress You Too Much!

Virtually everything in nature is correlated to some degree. Understanding this idea is extremely important! This means if you measure almost any variables you could conceive, you will necessarily find a correlation different from zero! Again, pause for a moment to consider how profound this statement is. It means if you go out into nature, take two empirically-derived variables at random, correlate them, **you will find a correlation in the sample.** It is a virtual guarantee, which begs the question why evaluating null hypotheses about a zero correlation in the population may seem a bit ridiculous (as many methodologists, not just myself, have long pointed out). What is more, with a large enough sample size, you can reject the null hypothesis for any correlation computed on a sample. Hence, when you see “ $p < 0.05$ ” for a correlation in a research report, it may (and often does) mean very little. You must look further into how large the correlation actually is and whether its statistical significance is mostly a function of sample size. That is, when you obtain a small correlation with your data, you must ask yourself when it is truly that “big of a deal” or if nothing really “is there” at all with your data. Often, researchers like to make “big deals” of very small correlations simply because the correlation comes out to be statistically significant, and these interpretations are often very much misguided if not downright dangerous. Can you imagine correlating a medical treatment for COVID-19 and concluding that the treatment works without having performed a proper experimental design? It is mandatory in this regard to understand the distinctions between a **test of correlation** vs. a **correlational design**. We discuss this in some detail later. Without a good design, you may obtain reasonably-sized correlations, but they may simply not be that meaningful due to profound differences between experimental and correlational research.

For now, the point is that all one can conclude from a statistically significant correlation is that there is evidence to suggest that the true correlation in the population from which the data were drawn is not equal to zero. That's it! A statistically significant correlation does not somehow imply, by itself, that the correlation is large, nor does it even imply the correlation is important. Substantively (read: scientifically), the correlation may be absolutely trivial, and it may even have been **a priori expected**, even if it is statistically significant. **There is nothing at all in nature that should presume that the correlation between variables we study should be zero. Most often, correlations are not zero, but that does not necessarily mean something meaningful or impressive is going on. It simply means most things are correlated.** Hence, when a researcher reports correlations and puts stars next to them to indicate statistical significance, you, as a critical consumer of research, should not automatically associate any measure of importance or even relevance to what has been reported. You simply need to know more about the problem at hand and under investigation to draw more precise conclusions. **The correlation itself is simply a mathematical abstraction.** The computation does not know anything about the “real” empirical objects you are applying the correlation to. This cannot be emphasized enough! What is more, even mediocre correlations still explain very little variance, a concept that will be elaborated on in our discussion of regression models later in the book. For example,

consider a correlation of 0.5. When squared, it explains only 25% of the variance. That is, when we square the Pearson correlation coefficient, we obtain a measure of the amount of variance explained:

$$r^2 = \text{proportion of variance explained}$$

So when a researcher simply claims a correlation between two variables, even if the correlation is quite moderate to large (e.g. 0.5) in absolute size, the total variance explained may be only 25%! Confident as they may be that they now understand a “link” between two variables, it is only the humble scientist who acknowledges that 75% of the variance is still a mystery. Indeed, $1 - r^2$ should be called the **coefficient of humility**. As a consumer, you will regularly come across “fascinating” research findings that nonetheless have quite modest effects. This is more to do with research scientists wanting to promote their results than anything necessarily impressive. The exaggeration is a **social phenomenon**, not a scientific one. As Einstein reminded us, even though we know “some” of what there is to know, most of what there is to be known we are still quite clueless about. Einstein was referring to the $1 - r^2$ idea. Marketing and emphasizing small findings should not be a part of science, but sadly it is for the purpose of generating careers (we all have to get paid somehow!). We expect this kind of thing from the popular press, but unfortunately marketing has also made its way into scientific journals for the purpose of career promotion. **Always be critical of what you are reading in even the best of media or journal publications. Is the claim that is being made representative of the actual research, or is at least some of it imbued with clever marketing making extraordinary claims that go far beyond the science?** Good journals (not necessarily always the most prestigious) publish good science, not merely “popular” science.

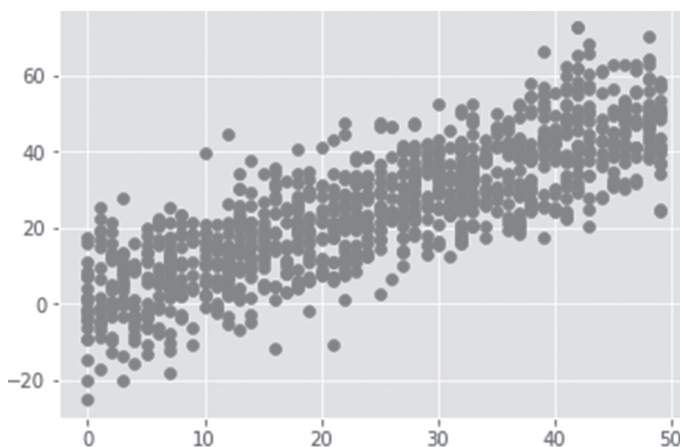
4.5 Computing Correlation in Python

Python allows us to easily compute a correlation coefficient on random data, and we do so to get us started. We can compute a correlation coefficient on random data via the following:

```
import numpy as np
np.random.seed(1)
x = np.random.randint(0, 50, 1000)
y = x + np.random.normal(0, 10, 1000)
np.corrcoef(x, y)
array([[1., 0.81543901],
       [0.81543901, 1.]])
```

For this data, we see that the correlation is equal to 0.815, a relatively strong correlation in absolute value. We can generate a scatterplot quite easily for these data using **matplotlib**:

```
import matplotlib
import matplotlib.pyplot as plt
%matplotlib inline
matplotlib.style.use('ggplot')
plt.scatter(x, y)
```



We see that the plot reveals the rather strong relationship between variables. There does not appear to be any visible bivariate outliers in the plot. For these data, a Pearson correlation coefficient is definitely appropriate because we have evidence of linearity in the plot. Even if the linearity were not present, had we theorized a linear relationship between the variables, then computing a Pearson correlation would still be well worth our time, even if to demonstrate the failure of our theory to agree with data.

We now consider covariance and correlation on a more elaborate data set. We consider the Galton data, which we obtained from R (only the first few cases are shown). In this data set are the heights of parents and their grown children. We would like to see if there might be a linear relationship between these two variables:

```
galton = pd.read_csv('Galton.csv')
```

```
Out[212]:
```

	Unnamed: 0	parent	child
0	1	70.5	61.7
1	2	68.5	61.7
2	3	65.5	61.7
3	4	64.5	61.7
4	5	64.0	61.7
5	6	67.5	62.2
6	7	67.5	62.2
7	8	67.5	62.2
8	9	66.5	62.2
9	10	66.5	62.2
10	11	66.5	62.2

```
galton.head()
```

```
Out[226]:
```

	Unnamed: 0	parent	child
0	1	70.5	61.7
1	2	68.5	61.7
2	3	65.5	61.7
3	4	64.5	61.7
4	5	64.0	61.7

```
from scipy import stats
pearson_coef, p_value = stats.pearsonr(galton["child"],
galton["parent"])
print("Pearson Correlation: ", pearson_coef, "and a P-value of:",
p_value)
```

```
Pearson Correlation: 0.45876236829282174 and a P-value of:
1.7325092920165045e-49
```

We can see from the above that the correlation between parent and child height is equal to 0.45, and has an exceedingly small p -value. Hence, it is statistically significant, and we can reject the null hypothesis that the true correlation in the population from which these data were drawn is equal to 0. Notice that the correlation here is between what Karl Pearson (famed statistician) would later call **organic variables** of heights. This was indeed one of the first uses of the correlation coefficient back around the year 1888. Galton and others wanted to show that generational characteristics and traits were transmitted genetically (heritability) from one generation to the next. Hence, a correlation here is extremely meaningful. That tall parents are more likely to have tall children (as adults) clues us into something potentially genetic going on across generations. Hence, though all the caveats earlier mentioned about interpreting correlations remains true (in that sometimes they might be quite meaningless), in cases such as with Galton's data, any correlation between generations was truly a meaningful result. The lesson is this: **what determines whether a correlation is meaningful or not is often a function of the variables on which it is computed, and not necessarily the size of the coefficient.** Therefore, the next time you are faced with a correlation coefficient, don't look at the absolute size of the number for a measure of importance. Rather, look at the variables on which it was computed to glean a sense of scientific meaning. If I told you there is a correlation of 0.2 between eating carrots and cancer, that correlation, though small numerically, would likely make you reconsider your vegetable choices. **The correlation is small numerically, but the pragmatic effect is huge.** Why? Because of the physical implications of the established relationship.

Notice that our null hypothesis above is that the correlation in the population is equal to 0. You might ask if that is a suitable null hypothesis, and you would be correct to ask that question. As we alluded earlier, why posit such a null hypothesis if it is virtually impossible in the first place that the correlation in the population could ever be equal to 0? Many excellent methodologists have asked the same question and have been critical about null hypotheses always being about such a particular value. As an example of this, the author once attended a presentation where the correlation between two variables was found to be approximately 0.2, and combined with the sufficiently large sample size, the null was, of course, rejected. The research scientist claimed a discovery! He essentially boasted about a correlation of 0.2. However, what he did not realize is that a correlation of approximately 0.2 for the variables he was working with was, in all likelihood, probably a reasonable **prior expectation** regardless of his scientific intervention. That is, obtaining a correlation of 0.2 was really "no big deal" at all, since even by chance this could have been anticipated. Since a correlation of 0 was automatically assumed under the null, the 0.2 was advertised as being due to his treatment. In all probability, his treatment had virtually zero effect and the correlation of 0.2 was status quo for his research variables to begin with. When you evaluate correlations and other measures blindly, software does not alert you that you are drawing

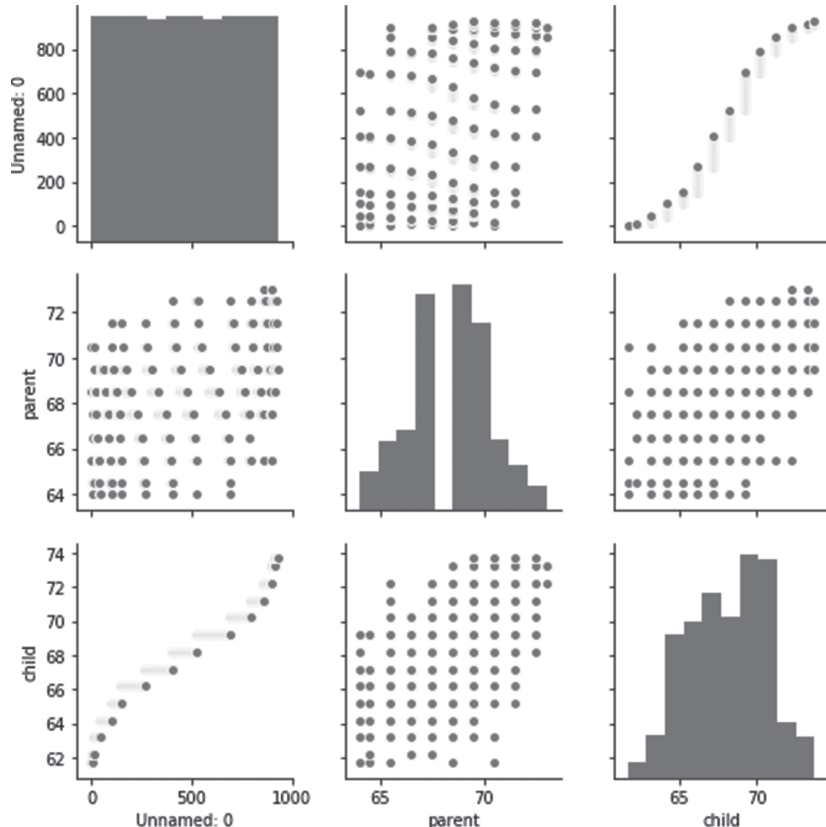
potentially meaningless conclusions. It simply computes things for you. **Always be critical about what null hypothesis is being advanced (and possibly rejected) when evaluating correlation coefficients. Rejecting a null hypothesis that the population coefficient is equal to 0 is usually not very impressive to begin with. You may still get it published, but it may still not be that big of a deal. That the prior expectation is set at 0 on a theoretical level is exactly that, theory, and is usually only for theoretical convenience for the statistical test (i.e. so a theoretical sampling distribution for the statistic can be easily obtained).**



Rejecting a null hypothesis that the population correlation coefficient is equal to 0 may not be a very impressive result, especially if a correlation greater than 0 would have naturally been expected due to chance anyway. Any statistical test can achieve the infamous “ $p < 0.05$ ” given a null hypothesis that is unrealistic. Hence, the statement “I found a statistically significant correlation” may not, from a scientific point of view, mean much at all. Always ask further questions and dig into the problem at hand to understand what is (or not) going on with the data you are studying.

We can obtain some plots of the correlation as well as histograms for both variables univariately through `sns.pairplot()`:

```
import seaborn as sns
sns.pairplot(galton)
Out[261]: <seaborn.axisgrid.PairGrid at 0xfb9dc18>
```



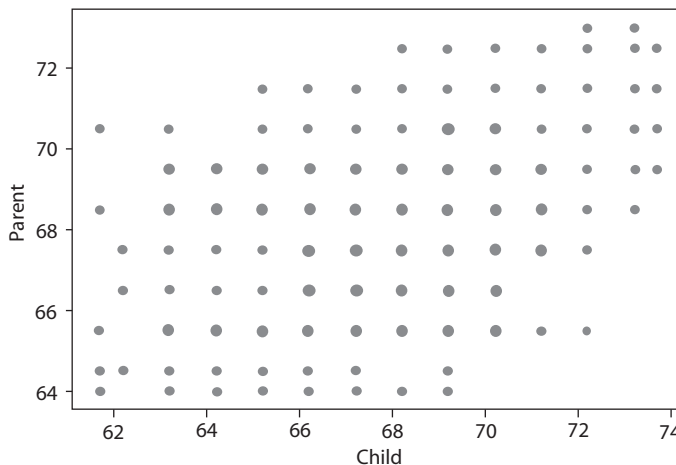
We can see that the correlation for parent and child is represented by the scatterplots in row 2, column 3, as well as row 3, column 2. The remaining scatterplots are of no use, as they simply represent the correlation of each variable with an index variable (“unnamed”) that Python produced automatically. Likewise, the plots in row 1, column 3, and row 3, column 1 are of no use. The histograms in row 2, column 2, and row 3, column 3 are useful, as they represent the distributions for parent and child, respectively. We can see that both distributions are relatively normal in shape, even if slightly skewed.

Instead of producing a matrix plot as earlier, we can instead define each variable from the Galton data (printing only a few cases on each variable), and generate a scatterplot through the following:

```
parent = galton['parent']
child = galton['child']
```

This extracts the variables that we will use in the scatterplot. We now build the plot using `plot.scatter()`:

```
columns = ['child', 'parent']
ax1 = galton.plot.scatter(x = 'child', y = 'parent')
```



An interesting point about scatterplots is that the data in them represent a subset of the **Cartesian product**. What is the Cartesian product? It is the total of all **possible pairs of points** in the plot. To understand this, imagine in this plot that all points were represented, that is, all possible pairings of child and parent. The correlation would then equal 0, because we would have the full Cartesian product. What the relationship between parent and child represents then is a **narrowing of this product**, such that a subset or subspace of the full Cartesian product set is represented. Indeed, this is precisely what defines a **mathematical relation**, which is a subset of the Cartesian product. For a good discussion of these ideas, see Hays (1994).

4.6 T-Tests for Comparing Means

T-tests are used in statistics for several reasons, one of which is as a **test for mean differences**. The reason why we say this is only one purpose for them is because *t*-tests are also used in assessing the statistical significance of regression coefficients

and other sample statistics in other contexts. That is, **they are not solely used to evaluate mean differences**. In introductory courses, often students automatically associate t -tests with mean differences. However, this is not the case. When we refer to t -statistics, we are referring to the t -distribution, but the application of this distribution is very widespread, not simply for assessing mean differences.

In a **one-sample t -test**, the researcher would like to evaluate the probability that an obtained sample mean could have reasonably been drawn from a particular population with a specified population mean. If the population variance for the problem is known, then performing a **z -test** instead of a **t -test** would be more appropriate. However, when the variance is unknown and has to be estimated based on sample data, then the t -test is the correct test to use. For relatively large sample sizes (e.g. greater than 100 or so), whether one performs the t -test or z -test will usually make little difference with regard to whether or not the null hypothesis is rejected. The reason for this is because as degrees of freedom get larger and larger, the t -distribution converges to that of a z -distribution. Still, to be on the safe side, you cannot go wrong with the t -test when variances are unknown and have to be estimated from sample data, regardless of how large your sample might be.

A closer look at the one-sample t -test reveals how it is constructed:

$$t = \frac{\bar{y} - E(\bar{y})}{\hat{\sigma}_M} = \frac{\bar{y} - E(\bar{y})}{s/\sqrt{n}}$$

where $\bar{y}$ is the sample mean we obtain from our research and $E(\bar{y})$ is the expectation under the null hypothesis, which is the population mean, μ , in this case. Hence, we could rewrite the numerator as $\bar{y} - \mu$. The denominator is the **estimated standard error of the mean**, where s is the sample standard deviation and n is the sample size. We see that the t -test is comparing a mean difference in the numerator to variation we would expect under the null in the denominator. We will explore the construction of the t -test in more detail in our discussion of statistical power in a later chapter. As we will see then, a large t does not necessarily imply an impressive scientific result, as a large t can be obtained in many cases by simply managing to lower s or increase n or both. The degrees of freedom for the one-sample t -test are equal to $n - 1$, that is, one less the number of participants in the sample.

We can easily demonstrate the one-sample t -test on some fictitious data. Suppose we would like to know whether it is reasonable to assume that the following IQ sample data could have been drawn from a population of IQ scores with mean equal to 100. Our null hypothesis is $H_0 : \mu = 100$ against the alternative hypothesis $H_1 : \mu \neq 100$. We first enter our data:

```
iq = [105, 98, 110, 105, 95]
df = pd.DataFrame(iq)
```

```
df
Out[38]:
0    105
1     98
2    110
3    105
4     95
```

We can conduct the one-sample t -test using **scipy**, listing first the dataframe (**df**), followed by the population mean under the null hypothesis, which for our data is equal to 100:

```
from scipy import stats
stats.ttest_1samp(df, 100.0)
```

```
Out[41]: Ttest_1sampResult(statistic=array([0.9649505]),
pvalue=array([0.38921348]))
```

The t -statistic for the test is equal to 0.9649505, with an associated p -value of 0.38921348. Since the p -value is rather large (definitely not less than some set level such as 0.05), we do not reject the null hypothesis. That is, we have insufficient evidence to suggest that our sample of IQ values was not drawn from a population with mean equal to 100. Notice that we have not “confirmed” that the population mean is equal to 100, only that we have failed to reject the null hypothesis. There is a difference here. A failure to reject the null never confirms any truth about the null; it merely gives us no reason to suggest it is false. It essentially means “nothing to see here, carry on.” If we fail to reject the null hypothesis that older folks get more COVID-19 than younger adults do, it does not mean they get the same amount. It simply means we failed to detect a difference. This distinction is important.

The **two-sample t -test** can be used when collecting data from what we assume, under the null, are two independent populations:

$$t = \frac{E(\bar{y}_1) - E(\bar{y}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{\mu_1 - \mu_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Notice that in the two-sample case, the mean difference in the numerator is between two population means, μ_1 and μ_2 . The standard error is now called the **estimated standard error of the difference in means**, and is composed of both sample variances from each sample as well as a corresponding sample size. We now conduct a two-sample t -test, this time using the previously loaded Galton data. We would like to evaluate the null hypothesis that the mean difference between parent and child is equal to 0:

```
from scipy import stats
stats.ttest_ind(parent, child)
```

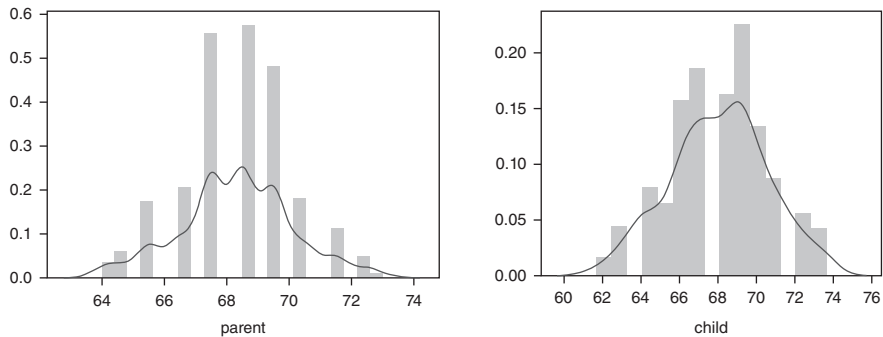
```
Out[278]: Ttest_indResult(statistic=2.167665371332533, pvalue=
0.030311049956448916)
```

The obtained t -statistic is equal to 2.167, with p -value of 0.03. Since the p -value is quite small (e.g. less than 0.05), we reject the null and conclude a mean population difference between parent and child. That is, we are rejecting the null hypothesis of $H_0 : \mu_1 = \mu_2$ in favor of the alternative hypothesis $H_1 : \mu_1 \neq \mu_2$. Be sure to note that these hypotheses are about population parameters, not sample statistics.

We can easily evaluate distributions for both variables by obtaining plots:

```
plt.figure(figsize=(10, 7))
sns.distplot(parent)

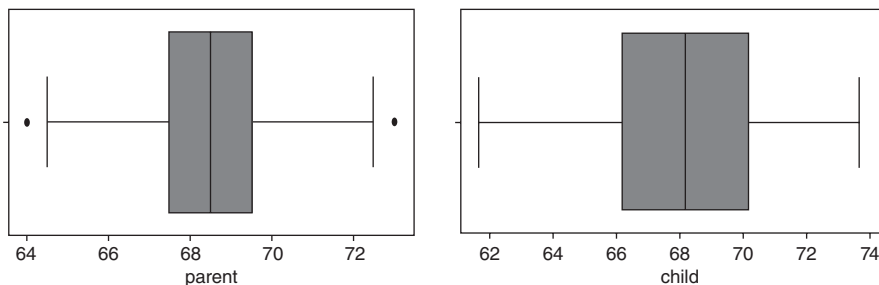
plt.figure(figsize=(9, 5))
sns.distplot(child)
```



We see from these plots that each population is relatively approximately normally distributed. Hence, we can say that one assumption for the *t*-test, that of **normality of populations**, is probably at least tentatively satisfied. For the *t*-test, we also require that the **variances** in each population are equal as well. Informally at least, the variances for each population also appear to be quite similar based on the plots, as the spread along the *x*-axis appears to be quite similar. However, if you try a **Levene test** on this data using `stats.levene(child, parent, center = 'mean')`, you will reject the null of equal variances. Levene's test is a test for the equality of variances in the population, though its *p*-value is also quite sensitive to sample size. Use these tests as a guide only.

We can also obtain **boxplots** for both the parent and child:

```
sns.boxplot(parent)
sns.boxplot(child)
```



We can see from these boxplots that parent might have a couple of rather extreme scores worth looking into. The center line in each plot represents the median, while the lines enclosing the rectangle are the first quartile (Q1) and the third quartile (Q3). We can see that the distribution for child is pulled ever so slightly to the left, indicating a mild negative skew. If you are not familiar with boxplots and would like a more informative overview, see Montgomery (2005).

4.7 Paired-Samples t-Test in Python

Instead of a one-sample or independent-samples situation, sometimes we have data measured on **pairs of observations**. A classic example is asking husbands and wives their happiness ratings for their marriage. The data cannot be presumed to be independent in this case. That is, knowing a husband’s rating likely tells us something about a wife’s rating (even if the ratings end up potentially being in different directions!). The paired situation is actually a special case of the more general **matched pairs design** where “blocks” of subjects are featured (Table 4.1).

In Table 4.1, each block represents either a natural pairing or one imposed by the researcher. In the case of the husband and wife pairing, “Block 1” would consist of the first married couple. Treatment 1 in this case might be the husband’s score, while treatment 2 might be the wife’s score. Alternatively, in a different set-up, perhaps treatment 1 is one drug dose and treatment 2 another, and so on. We can see that the matched pairs and block layout is quite general and can accommodate a variety of specific experiments. The key point again is that **within each block, there is a lack of independence between conditions**. In a purely between-subjects design, it is assumed that observations under both conditions are independent. If we generalize this situation to where there are more than two conditions, we obtain results given in Table 4.2, where there are several conditions instead of only two.

We will discuss these designs later in the book in a bit more detail when we feature repeated-measures ANOVA. As we will see, repeated-measures ANOVA is nothing more than a generalization of the randomized block design.

Table 4.1 Matched design.

	Treatment 1	Treatment 2
Block 1	10	8
Block 2	15	12
Block 3	20	14
Block 4	22	15
Block 5	25	24

Table 4.2 Randomized block design.

	Treatment 1	Treatment 2	Treatment 3
Block 1	10	9	8
Block 2	15	13	12
Block 3	20	18	14
Block 4	22	17	15
Block 5	25	25	24

Table 4.3 Learning as a function of trial (hypothetical data).

Rat	Trial			Rat Means
	1	2	3	
1	10.0	8.2	5.3	7.83
2	12.1	11.2	9.1	10.80
3	9.2	8.1	4.6	7.30
4	11.6	10.5	8.1	10.07
5	8.3	7.6	5.5	7.13
6	10.5	9.5	8.1	9.37
Trial means	$M = 10.28$	$M = 9.18$	$M = 6.78$	

Returning to the situation where there are only two conditions, but that they are related by their pairing, a **paired-samples *t*-test** is appropriate. We demonstrate the test on the data given in Table 4.3, in which rats were measured on a learning task across three trials.

Notice that for these data, each rat is measured a total of three times from trials 1 through 3. For our purposes, because we are discussing the paired situation only, we will focus only on the first two trials. Since a rat serving under trial 1 also serves under trial 2, the data are **naturally paired**, and actually, in this situation, are **repeated**. For these data, knowing the behavior of one rat under one trial tells us information about how that same rat might behave under another trial. Hence, we have no reason to believe the trials are independent. We first build the dataframe for trials 1 and 2:

```
trial_1 = [10, 12.1, 9.2, 11.6, 8.3, 10.5]
trial_2 = [8.2, 11.2, 8.1, 10.5, 7.6, 9.5]
paired_data = pd.DataFrame(trial_1, trial_2)
paired_data
```

```
Out[66]:
8.2    10.0
11.2   12.1
8.1     9.2
10.5   11.6
7.6     8.3
9.5    10.5
```

We now conduct the *t*-test:

```
stats.ttest_rel(trial_1, trial_2)
Out[69]: Ttest_relResult(statistic=7.201190377787752, pvalue=
0.0008044382024663002)
```

The value of the *t*-statistic is equal to 7.20, with an associated *p*-value of 0.0008, which is statistically significant. Therefore, we have evidence to suggest that the mean difference between the two trials is not equal to 0. That is, we reject the null hypothesis and

infer the alternative hypothesis that in the population from which these data were drawn, the true mean difference is unequal to 0. Again, we will return to our discussion of paired data when we discuss blocking designs and repeated measures models more thoroughly later in the book.

4.8 Binomial Test

The binomial distribution is useful for modeling the probability of success on a stochastic experiment that is **binary** and whose events are **mutually exclusive**. The classic example of where the binomial can be used is in coin-flipping, where the experiment or trial can result in either a head or tail. It is a binary event since the only possibilities are one of two categories when the experiment is performed. Other examples include being dead or alive, day or night, young or old, etc. The **binomial distribution** is given by

$$\begin{aligned} p(r) &= \binom{n}{r} p^r (1-p)^{n-r} \\ &= \left(\frac{n!}{r!(n-r)!} \right) p^r (1-p)^{n-r} \end{aligned}$$

where $p(r)$, read as “probability of r ,” is the probability of observing r successes. The total number of trials is denoted by n . Hence, $p(r)$ is the probability of observing a given number of successes out of a total number of trials (e.g. number of coin flips). Since the events in question are mutually exclusive, the total probability is equal to $p + (1-p)$, where $1-p$ is often denoted by q . Hence, $p + (1-p) = p + q = 1$. Returning to our example of how the binomial distribution can be used to answer a research question, let us unpack this a bit further. Suppose we were interested in learning of the probability of a given number of heads on a presumed fair coin. We set $p = 0.5$ to denote the **assumed** probability of heads under the null. Suppose we obtain 2 heads out of 5 flips, and we would like to know the probability of obtaining 2 or more heads assuming the coin is fair. This is a binomial setting since:

- The coin variable is **binary** in nature, meaning that the outcome can only be one of two possibilities, “head” or “tail.” It is a **discrete variable**. The two events are mutually exclusive, meaning that you either get a head or a tail, and cannot get both on the same flip.
- The probability of “success” on each trial remains the same from trial to trial. That is, the probability is **stationary**. In other words, if the probability of heads on the first trial is 0.5, then this implies the probability of heads on the second and ensuing trials is also 0.5.
- Each trial is **independent** of any other trial. For instance, the probability of a “head” on the first flip does not have an influence on the probability of heads on the second flip, and so on.

These conditions make the binomial distribution suitable for modeling the probability of success (in this case, “heads”) on the coin. In Python, we can evaluate the

probability of 2 or more heads out of 5 flips as follows, where `stats.binom_test()` is the function for the test, “2” is the number of successes out of `n=5` trials, and `p=0.5` is the probability of a success on any given trial:

```
stats.binom_test(2, n=5, p=0.5, alternative='greater')
Out[71]: 0.8125
```

Hence, we see that the probability of obtaining 2 or more heads on 5 flips of a fair coin is equal to 0.8125. But what can we make of this result? Well, if we started out by assuming the coin was fair, where the probability of heads on any given flip were equal to 0.5, then a finding that the probability of 2 heads out of 5 flips were equal to 0.8125 would likely not give us evidence to reject the null hypothesis. That is, what the result is telling us is that on fair coins, **2 heads out of 5 flips happens quite a lot**. It is a common occurrence, assuming we believe in probability theory at all (if we do not, well then, that’s a different story). Hence, it gives us no reason to doubt the assumption that the coin is fair. The probability of the data given the null is relatively high.

4.9 The Chi-Squared Distribution and Goodness-of-Fit Test

The binomial test just surveyed for applications can actually be considered a special case of an even more popular test in statistics, that of the **chi-squared goodness-of-fit test**. The chi-squared distribution is not necessarily equivalent to a chi-squared “test” and the two should not be unequivocally equated. In undergraduate applied statistics, chi-square is simply associated with the chi-squared test, but chi-square, like *t*, is simply a statistic with a proper density that can be used in different scenarios to evaluate null hypotheses. In theoretical statistics, the chi-square density pops up virtually everywhere and is associated with the *F* distribution and others.

The **chi-square distribution** is given by

$$f(x) = \frac{1}{2^{v/2} \Gamma(v/2)} x^{[(v/2)-1]} e^{-x/2}$$

where for different inputs of $x > 0$, v are degrees of freedom and Γ is the **gamma function**. We can also state the chi-square distribution in terms of the sum of squares of n independent and normally distributed *z*-scores:

$$\chi_n^2 = \sum_{i=1}^n z_i^2 = \sum_{i=1}^n \frac{(x_i - \mu)^2}{\sigma^2}$$

The **goodness-of-fit test** happens to be a statistical method that features the chi-square distribution in evaluating the statistical significance of the test. The goodness-of-fit test features a test on **count data** rather than means such as in the *t*-test or ANOVA. We define the chi-squared goodness-of-fit test as follows:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c (O_i - E_i)^2 / E_i$$

where O_i are observed frequencies and E_i are expected frequencies within each cell. The summation notation denotes summing these squared cell differences, $(O_i - E_i)^2$ within c columns and then across r rows. Notice that χ^2 takes squared differences in the numerator for the reason that if we did not square these, the sum would equal 0 and hence the resulting value for χ^2 would always equal 0 regardless of the discrepancy between O_i and E_i . The act of squaring discrepancies or deviations serves to ensure a value for χ^2 different from zero so long as there is at least some discrepancy between observed and expected differences. If the sum of $(O_i - E_i)^2$ is large relative to E_i , then, given the degrees of freedom, χ^2 may become extreme and attain statistical significance. **The null hypothesis then in such a test is that observed frequencies are equivalent to those expected in the population from which the data were drawn.** The alternative hypothesis is that they are not, or equivalently, that there is some **association** between the row variable and the column variable.

Performing a chi-squared goodness-of-fit test in Python is very easy. As an example, let us first consider the case where all observed frequencies are equal in each cell. In the following, we set them all equal to 16:

```
from scipy.stats import chisquare
chisquare([16, 16, 16, 16, 16])

Out[74]: Power_divergenceResult(statistic=0.0, pvalue=1.0)
```

We see that the p -value comes out to be 1.0, for the reason that when all cell frequencies are equal, there can be no discrepancy between those observed and expected. In other words, the result we are witnessing is the exact result we would expect under the null hypothesis. In this case, the total frequencies available are $16(5) = 80$. In this **one-way** (“one-way” because we only have one variable) chi-squared analysis, if there is no discrepancy between cells, then this total frequency should distribute itself evenly across the cells, which is exactly what the above suggests. There is a total of 16 cases per cell, all cells equal, and hence there is absolutely no evidence against the null hypothesis, which is why the p -value is equal to 1.0.

Now we demonstrate the situation where frequencies are slightly **unequal** by adjusting a couple of the cell frequencies downward to 15. Notice elements 2 and 4 in our data in the following are now data points 15 rather than 16:

```
chisquare([16, 15, 16, 15, 16])

Out[75]: Power_divergenceResult(statistic=0.07692307692307693,
pvalue=0.9992790495310748)
```

For these data, we see the resulting p -value is now equal to 0.999, a bit lower than 1.0. This is as a result of the discrepancy in frequencies now from cell to cell. That is, the probability of observing this set of frequencies under the null hypothesis of equal frequencies is still very high, certainly no cause or reason to reject the null hypothesis. The following is an even more disparate distribution of frequencies:

```
chisquare([16, 15, 10, 8, 25])

Out[76]: Power_divergenceResult(statistic=11.81081081081081, pvalue=0.01881499899397827)
```

We can see in this case that the observed p -value is equal to 0.018, and hence we deem the result to be statistically significant at the 0.05 level. That is, the probability of observing a discrepancy such as we have observed is very small under the null hypothesis of equal frequencies per cell. Hence, we reject the null hypothesis of equal frequencies per cell.

4.10 Contingency Tables

We just considered the case of the one-way goodness-of-fit test, where we had only a single row. However, the formulation $\chi^2 = \sum_{i=1}^r \sum_{j=1}^c (O_i - E_i)^2 / E_i$, as mentioned, allows us to generalize to a contingency table with rows and columns. In most practical cases, of course, we have more than a single variable on which we have frequencies. Often research provides us with a contingency table where we have at least one column and one row variable, such as in Table 4.4.

In the data in Table 4.4, the column variable is whether or not a **condition** is present. For this example, the condition can be anything we like. For example, suppose the condition variable is that of whether or not symptoms of post-traumatic stress disorder are present and the row variable of exposure is whether an individual has been exposed or not exposed to combat in Iraq. Or, in the age of COVID-19, the condition could be whether symptoms of the disease are present and the exposure variable that of whether an individual has been exposed via contact tracing (i.e. a measure of whether the given individual has been in contact with those having already been confirmed for the illness). For the sake of example, it does not really matter what we name these variables. The null hypothesis for these data can be framed in at least a couple of ways. One way of stating it is to say that the 50 counts making up the table are essentially randomly distributed across the cells (with consideration of the row and column marginal totals in computing expectations per cell, to be discussed shortly) and that there is no systematic pattern to them. If there were a systematic pattern, it might suggest some cells have disproportionate numbers compared to other cells (again, relative to expectations computed using marginal row and column totals), which would suggest an **association** between the column and row variables.

For example, the expectation for the cell in row 1, column 1 would be computed as the row total of 30 times the column total of 25, divided by the total sample size of 50, yielding an expectation of 15 for the first cell. We see that the observed value of 20 for that cell deviates from 15. The chi-squared analysis will repeat these computations per cell and tally up the squared deviations. If overall observed frequencies deviate from

Table 4.4 Contingency table for a 2×2 design.

	Condition Present (1)	Condition Absent (0)	Total
Exposure yes (1)	20	10	30
Exposure no (2)	5	15	20
Total	25	25	50

those expected, this suggests an association between variables. Hence, this is a second way of stating the null, that there is no association between the two variables. Notice that this is analogous to the one-way goodness-of-fit test, only that now we have both a row and a column variable. Otherwise, the concept is exactly the same. To evaluate the null, we will compute a chi-squared statistic and evaluate it for statistical significance.

In Python, we set up our contingency data as follows:

```
import numpy as np
matrix = np.array([[20, 10],
                   [5, 15]])

matrix
Out[23]:
array([[20, 10],
       [ 5, 15]])
```

To accommodate the test in `scipy.stats()` we organize the data in the following way, and also import `chi2_contingency()` to run the test:

```
from scipy.stats import chi2_contingency
obs = np.array([[20, 10], [5, 15]])

chi2_contingency(obs)
Out[21]:
(6.75, 0.0093747684594349, 1, array([[15., 15.],
                                     [10., 10.]])
```

The observed chi-squared statistic is equal to 6.75, with an associated p -value of 0.009. Since the p -value is quite small, we reject the null hypothesis of no association between the row and column variable and conclude an association between these two variables.

Review Exercises

1. Discuss the nature of a **Pearson Product-Moment correlation coefficient**. What does it seek to measure, exactly?
2. Why should a **plot** of data always accompany the reporting of a correlation coefficient? What is the danger in not including such a plot?
3. Demonstrate with some fictional data why computing the **covariance** is insufficient as a measure of the relationship between two variables. Why is it not dimension-free? Why is computing the correlation the solution to this problem?
4. When should **Spearman's rho** correlation be used in place of that of Pearson? Why might the two computations not agree with one another? Explain.
5. If “**everything is correlated to some degree**,” as claimed in the chapter, why bother then with evaluating a null hypothesis that the population correlation is equal to 0?
6. A researcher claims a **scientific discovery** has been made since he computed a correlation of $r = 0.20$ between his variables. What kinds of questions would you ask the researcher in critically evaluating such a claim?

7. Why might rejecting a **null hypothesis** in a test of a correlation coefficient not be that meaningful?
8. Does increasing **sample size** in the computation of Pearson r have any effect on the size of r ? Why or why not? Explain.
9. Consider the **one-sample t -test** for a mean. How might a large difference in the numerator still not result in a statistically significant value for t ? Explain.
10. A scientist claims, “I rejected the null hypothesis for my one-sample t -test.” What kinds of follow-up questions would you ask next?
11. Demonstrate with made-up data why if you change the null hypothesis for a one-sample t -test, the decision on the null will change even if the data remain exactly the same. Use Python to demonstrate your computations.
12. What are the **degrees of freedom** for the two-sample t -test? Why are they equal to such? Look at the layout for a two-sample case and see if you can figure out why those degrees of freedom are equal to what they are.
13. How is a **paired-samples t -test** different from a **two-sample t -test**? Explain.
14. Compute in Python the probability of obtaining 20 heads on 50 flips of a fair coin. What does your result suggest about the coin? Does it “prove” anything about the coin?
15. Explain how the **chi-squared goodness-of-fit test** statistic is similar in some ways to the t -test computation. What are the parallels? How are they different?